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### Notification method, notification system, and vehicle

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#### Abstract

A notification method for giving a notification to a driver sitting in a driver's seat of a vehicle includes: setting a target level of strength of the notification; when the target level is a first level, giving a visual notification from a visual device of the vehicle without giving a sound notification from a speaker of the vehicle; and when the target level is higher than the first level, giving a strong visual notification from the visual device or the sound notification from the speaker. The strong visual notification is stronger than the visual notification that is given when the target level is the first level.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-187518 filed on Nov. 24, 2022, incorporated herein by reference in its entirety.

### BACKGROUND

1. Technical Field

(2) The present disclosure relates to notification methods, notification systems, and vehicles.

2. Description of Related Art

(3) Japanese Unexamined Patent Application Publication No. 2019-159711 (JP 2019-159711 A) discloses an alertness maintaining device for maintaining the level of alertness of a driver of a vehicle. When a decrease in level of alertness of the driver is detected, the alertness maintaining device gives a sound stimulus of intensity corresponding to the level of alertness to the driver.

### SUMMARY

(4) According to the technique disclosed in JP 2019-159711 A, a sound stimulus is given to the driver of the vehicle. Sound stimuli are considered to be intense and effective. However, because of their intensity, drivers feel that the sound stimuli are too much. That is, drivers feel annoyed by the sound stimuli.

(5) The present disclosure provides a technique capable of giving a notification of appropriate strength to a driver of a vehicle.

(6) A first aspect of the present disclosure is a notification method for giving a notification to a driver sitting in a driver's seat of a vehicle. The notification method includes: setting a target level of strength of the notification; when the target level is a first level giving a visual notification from a visual device of the vehicle without giving a sound notification from a speaker of the vehicle; and when the target level is higher than the first level, giving a strong visual notification from the visual device or the sound notification from the speaker. The strong visual notification is stronger than the visual notification that is given when the target level is the first level.

(7) In the first aspect, when the target level is the first level, the sound notification may not be given from the speaker and a first visual notification may be given from a first visual device. The visual device may include the first visual device. When the target level is a second level, the sound notification may not be given from the speaker and a second visual notification may be given from a second visual device. The visual device may include the second visual device. The second level may be higher than the first level. A first angle between a first direction and a forward direction may be greater than a second angle between a second direction and the forward direction. The forward direction may be a direction from the driver's seat toward front of the vehicle, and the first direction may be a direction from the driver's seat toward the first visual device. The second direction may be a direction from the driver's seat toward the second visual device.

(8) In the first aspect, the first angle may be greater than a reference angle and may be smaller than a limit angle. The reference angle may be half an angle of a human's average effective visual field, and the limit angle may be half an angle of a human's average visual field. The second angle may be smaller than the reference angle.

(9) In the first aspect, when the target level is a third level, the sound notification may be given from the speaker. The third level may be higher than the second level.

(10) In the first aspect, the notification method may further include: determining whether the notification to the driver is necessary; and determining whether the driver reacts to the visual notification when the target level is the first level. When the notification to the driver is necessary,

the target level may be initialized to the first level. When the driver does not react to the visual notification, the target level may be set to a level higher than the first level.

(11) In the first aspect, the notification method may further include: determining whether the notification to the driver is necessary; determining whether the driver reacts to the first visual notification; and determining whether the driver reacts to the second visual notification. When the notification to the driver is necessary, the target level may be initialized to the first level. When the driver does not react to the first visual notification, the target level may be set to the second level. When the driver does not react to the second visual notification, the target level may be set to the third level.

(12) A second aspect of the present disclosure is a notification system that gives a notification to a driver sitting in a driver's seat of a vehicle. The notification system includes one or more processors. The one or more processors are configured to set a target level of strength of the notification. The one or more processors are configured to, when the target level is a first level, give a visual notification from a visual device of the vehicle without giving a sound notification from a speaker of the vehicle. The one or more processors are configured to, when the target level is higher than the first level, give a strong visual notification from the visual device or the sound notification from the speaker. The strong visual notification is stronger than the visual notification that is given when the target level is the first level.

(13) In the second aspect, a first angle between a first direction and a forward direction may be greater than a second angle between a second direction and the forward direction. The forward direction may be a direction from the driver's seat toward front of the vehicle. The first direction may be a direction from the driver's seat toward a first visual device. The second direction may be a direction from the driver's seat toward a second visual device. The one or more processors may be configured to, when the target level is the first level, give a first visual notification from the first visual device without giving the sound notification from the speaker. The visual device may include the first visual device. The one or more processors may be configured to, when the target level is a second level, give a second visual notification from the second visual device without giving the sound notification from the speaker. The visual device may include the second visual device. The second level may be higher than the first level.

(14) A third aspect of the present disclosure is a vehicle. The vehicle includes a visual device, a speaker, and a control device that is configured to control a notification to a driver sitting in a driver's seat. The control device is configured to set a target level of strength of the notification. The control device is configured to, when the target level is a first level, give a visual notification from the visual device without giving a sound notification from the speaker. The control device is configured to, when the target level is higher than the first level, give a strong visual notification from the visual device or the sound notification from the speaker. The strong visual notification is stronger than the visual notification when that is given the target level is the first level.

(15) In the third aspect, a first angle between a first direction and a forward direction may be greater than a second angle between a second direction and the forward direction. The forward direction may be a direction from the driver's seat toward front of the vehicle. The first direction may be a direction from the driver's seat toward a first visual device, and the second direction may be a direction from the driver's seat toward a second visual device. The control device may be configured to, when the target level is the first level, give a first visual notification from the first visual device without giving the sound notification from the speaker. The visual device may include the first visual device. The control device may be configured to, when the target level is a second level, give a second visual notification from the second visual device without giving the sound notification from the speaker. The visual device may include the second visual device. The second level may be higher than the first level.

(16) According to the present disclosure, when the target level of the strength of the notification is low, no sound notification is given and a visual notification is given. A stimulus that is given to the

driver by the visual notification is weaker than a stimulus that is given to the driver by the sound notification. Therefore, the driver is less likely to feel annoyed by the notification.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
- (2) FIG. 1 is a conceptual diagram illustrating an overview of a notification system according to an embodiment;
- (3) FIG. 2 is a block diagram showing an example of the configuration of the notification system according to the embodiment;
- (4) FIG. 3 shows an example of a plurality of visual devices mounted on a vehicle according to the embodiment;
- (5) FIG. 4 is a conceptual diagram illustrating the positional relationship between a first visual device and a second visual device according to the embodiment;
- (6) FIG. 5 is a conceptual diagram illustrating an overview of a notification control process according to the embodiment;
- (7) FIG. 6 is a flowchart showing a first example of the notification control process according to the embodiment;
- (8) FIG. 7 is a flowchart showing a second example of the notification control process according to the embodiment;
- (9) FIG. 8 is a conceptual diagram showing various patterns of the notification control process according to the embodiment; and
- (10) FIG. 9 is a block diagram showing an example of the configuration of a vehicle control system according to the embodiment.

### DETAILED DESCRIPTION OF EMBODIMENTS

- (11) An embodiment of the present disclosure will be described with reference to the accompanying drawings.

#### 1. Notification System

- (12) FIG. 1 is a conceptual diagram illustrating an overview of a notification system 10 according to the present embodiment. The notification system 10 is mounted on a vehicle 1 and gives various notifications N to a driver 3 sitting in a driver's seat 2 of the vehicle 1. The vehicle 1 may be an autonomous vehicle. An operator sitting in the driver's seat 2, even during autonomous driving, is herein referred to as "driver 3."

- (13) The notification system 10 gives a notification N when a predetermined notification condition is satisfied. For example, the notification condition is that the level of alertness of the driver 3 falls below a threshold. As another example, the notification condition is that it becomes necessary to request the driver 3 to monitor the surroundings during autonomous driving of the vehicle 1. As still another example, the notification condition is that it becomes necessary to request the driver 3 to hold a steering wheel during autonomous driving of the vehicle 1. As yet another example, the notification condition is that it becomes necessary to request the driver 3 to switch to manual driving during autonomous driving of the vehicle 1. As a further example, the notification condition is that it becomes necessary to request the driver 3 to perform a braking operation. As a still further example, the notification condition is that driver assistance control for assisting with driving of the vehicle 1 is activated. Examples of the driver assistance control include collision avoidance control and lane departure prevention control.

- (14) Types of notifications N to the driver 3 include "visual notifications NV" and "sound

notifications NA.” The visual notifications NV are notifications using visual information such as light emission and display. The sound notifications NA are notifications using sound information such as a voice message and a beeping sound.

(15) FIG. 2 is a block diagram showing an example of the configuration of the notification system **10** according to the embodiment. The notification system **10** includes a visual device **20**, a speaker **30**, a driver monitor **40**, and a control device **50**.

(16) The visual device **20** gives visual notifications NV to the driver **3**. Examples of the visual device **20** include a light-emitting device and a display panel. The light-emitting device is, for example, a light-emitting diode (LED). The light-emitting device can provide (convey) different kinds of information by emitting light in different colors. The display panel is, for example, a liquid crystal panel or an organic electroluminescence (EL) panel. The visual device **20** is typically installed inside the cabin of the vehicle **1**.

(17) The speaker **30** gives sound notifications NA to the driver **3**. The speaker **30** is typically installed inside the cabin of the vehicle **1**.

(18) The driver monitor **40** is a device for detecting the state and behavior of the driver **3**. For example, the driver monitor **40** includes a camera installed at a position where the camera can capture images of the driver **3**. The driver monitor **40** may include a steering touch sensor installed on the steering wheel that is operated by the driver **3**. The driver monitor **40** may detect driving operations of the driver **3**.

(19) The control device **50** controls the notification system **10**. More specifically, the control device **50** controls notifications N to the driver **3** by controlling the visual device **20** and the speaker **30**. The control device **50** also analyzes information detected by the driver monitor **40**. For example, the control device **50** detects various states of the driver **3** by analyzing images captured by the camera. For example, the control device **50** can detect the face orientation, line of sight, eyes opening degree, mouth opening degree, etc. of the driver **3**. The control device **50** can also determine whether the driver **3** is holding the steering wheel or has his or her hands off the steering wheel, based on the detection result from the steering touch sensor.

(20) The control device **50** includes one or more processors **51** (hereinafter simply referred to as “processor **51**”) and one or more storage devices **52** (hereinafter simply referred to as “storage device **52**”). The processor **51** performs various processes. The storage device **52** stores various kinds of information. Examples of the storage device **52** include a volatile memory, a nonvolatile memory, a hard disk drive (HDD), and a solid state drive (SSD). The processor **51** may execute a notification control program that is a computer program for controlling the notification system **10**. The functions of the control device **50** are implemented through cooperation between the processor **51** that executes the notification control program and the storage device **52**. The notification control program is stored in the storage device **52**. Alternatively, the notification control program may be recorded on a computer-readable recording medium.

(21) The notification system **10** may include a plurality of the visual devices **20**. For example, the notification system **10** includes a first visual device **20-1** and a second visual device **20-2**. The first visual device **20-1** gives a first visual notification NV1 to the driver **3**. The second visual device **20-2** gives a second visual notification NV2 to the driver **3**. As shown in FIG. 2, the visual device **20** may include the first visual device **20-1** and the second visual device **20-2**.

(22) FIG. 3 shows an example of the visual devices **20** mounted on the vehicle **1**. The visual devices **20** include a door illumination **21** installed on a door, a pillar illumination **22** installed on a window frame, an instrument panel illumination **23** installed on an instrument panel, and a steering illumination **24** installed on the steering wheel. Each illumination is a light-emitting device including a plurality of LEDs. Each illumination is in a strip shape.

(23) A forward direction **D0** is a direction from the driver's seat **2** toward the front of the vehicle **1**. In the example shown in FIG. 3, a door illumination **21L**, a pillar illumination **22L**, and an instrument panel illumination **23L** are installed on the left side of the forward direction **D0**. A door

illumination **21R**, a pillar illumination **22R**, and an instrument panel illumination **23R** are installed on the right side of the forward direction **D0**. The steering illumination **24** is installed substantially on the forward direction **D0**.

(24) FIG. **4** is a conceptual diagram illustrating the positional relationship between the first visual device **20-1** and the second visual device **20-2** according to the present embodiment. The forward direction **D0** is a direction from the driver's seat **2** toward the front of the vehicle **1**. A first direction **D1** is a direction from the driver's seat **2** toward the first visual device **20-1**. A second direction **D2** is a direction from the driver's seat **2** toward the second visual device **20-2**. A first angle  $\theta 1$  is an angle between the first direction **D1** and the forward direction **D0**. A second angle  $\theta 2$  is an angle between the second direction **D2** and the forward direction **D0**. As shown in FIG. **4**, the first angle  $\theta 1$  is greater than the second angle  $\theta 2$ . Therefore, when the driver **3** sitting in the driver's seat **2** is looking straight ahead (forward direction **D0**), the first visual device **20-1** is far from the gaze direction of the driver **3**, and the second visual device **20-2** is close to the gaze direction of the driver **3**.

(25) A human's visual field is divided into an effective visual field (central visual field) and a peripheral visual field. Humans can clearly perceive objects in their effective visual field. On the other hand, humans can vaguely perceive objects in their peripheral visual field. Preferably, when the driver **3** sitting in the driver's seat **2** is looking straight ahead (forward direction **D0**), the first visual device **20-1** is in the peripheral visual field of the driver **3**, and the second visual device **20-2** is in the central visual field of the driver **3**.

(26) The angle of a human's average visual field will be described. The angle of the human's average visual field is the average value of the angles of the visual fields of a certain number of humans. For example, the human's average visual field extends about 100 degrees laterally (total of about 200 degrees), and a human's average effective visual field extends about 30 degrees laterally (total of about 60 degrees). A reference angle  $\alpha$  is half the angle of the human's average effective visual field (e.g., 30 degrees). A limit angle  $\beta$  is half the human's average visual field (e.g., 100 degrees). The first angle  $\theta 1$  between the first direction **D1** and the forward direction **D0** may be greater than the reference angle  $\alpha$  and smaller than the limit angle  $\beta$ . The second angle  $\theta 2$  between the second direction **D2** and the forward direction **D0** may be smaller than the reference angle  $\alpha$ .

(27) In the example shown in FIG. **3** described above, there is a possibility that the door illumination **21L**, the pillar illumination **22L**, the instrument panel illumination **23L**, and the door illumination **21R** may belong to the first visual device **20-1**. There is also a possibility that the steering illumination **24** may belong to the second visual device **20-2**.

## 2. Notification Control Process

(28) A stimulus that is given to the driver **3** by a sound notification **NA** is more intense than a stimulus that is given to the driver **3** by a visual notification **NV**. Sound notifications **NA** are effective in that sense. However, due to the intense stimuli of the sound notifications **NA**, the driver **3** may feel that the sound notifications **NA** are too much. That is, the driver **3** may feel annoyed by the sound notifications **NA**. Also, the sound notifications **NA** can be heard not only by the driver **3** but also by passengers. Therefore, the passengers may also feel annoyed by the sound notifications **NA**.

(29) It is therefore desired to give notifications **N** of appropriate strength to the driver **3** of the vehicle **1**. Therefore, the notification system **10** (control device **50**) according to the present embodiment performs a "notification control process" as described below.

(30) FIG. **5** is a conceptual diagram illustrating an overview of the notification control process according to the present embodiment. In the following description, the "strength of a notification **N**" means the "intensity of a stimulus that is given to the driver **3** by the notification **N**." The sound notifications **NA** from the speaker **30** are stronger than the visual notifications **NV** from the visual device **20**. In other words, the visual notifications **NV** from the visual device **20** are weaker than the sound notifications **NA** from the speaker **30**. Also, of the visual notifications **NV**, the second

visual notification NV2 from the second visual device **20-2** is stronger than the first visual notification NV1 from the first visual device **20-1**. In other words, the first visual notification NV1 from the first visual device **20-1** is weaker than the second visual notification NV2 from the second visual device **20-2**.

(31) The control device **50** sets (determines) a “target level” of the strength of a notification N. The control device **50** then determines the type of notification N to the driver **3** according to the target level. When the target level is relatively low, the control device **50** selects a relatively weak notification N. Specifically, when the target level is relatively low, the control device **50** does not give a sound notification NA from the speaker **30** and gives a visual notification NV from the visual device **20**. When the target level is relatively high, the control device **50** selects a relatively strong notification N. Specifically, when the target level is relatively high, the control device **50** gives a stronger visual notification NV from the visual device **20** or gives a sound notification NA from the speaker **30**.

(32) For example, the target level includes the following three options: a first level LV1, a second level LV2, and a third level LV3. The second level LV2 is higher than the first level LV1, and the third level LV3 is higher than the second level LV2. When the target level is the first level LV1, no strong notification N is necessary, and the first visual notification NV1 is selected. When the target level is the third level LV3, a strong notification N is necessary, and a sound notification NA is selected. When the target level is the second level LV2, the second visual notification NV2 of intermediate strength is selected.

#### Effects

(33) As described above, when the target level of the strength of a notification N is low, no sound notification NA is given but a visual notification NV is given. A stimulus that is given to the driver **3** by a visual notification NV is weaker than a stimulus that is given to the driver **3** by a sound notification NA. Therefore, the driver **3** is less likely to feel annoyed by the notification N. Moreover, since no sound notification NA is given, the passengers are less likely to feel annoyed by the notification N.

(34) The visual notifications NV may include the first visual notification NV1 that is relatively weak and the second visual notification NV2 that is relatively strong. When the target level of the strength of a notification N is the first level LV1 that is very low, the first visual notification NV1 is selected. When the target level is the second level LV2 higher than the first level LV1, the second visual notification NV2 is selected. This provides finer notification control. That is, when the target level is the first level LV1 that is very low, the possibility of the driver **3** feeling annoyed by the notification N is effectively reduced. When the target level is the second level LV2 higher than the first level LV1, the attention of the driver **3** can be drawn to some extent while reducing the annoyance the driver **3** feels.

(35) When the target level of the strength of a notification N is high, a sound notification NA is given. Since the sound notification NA gives an intense stimulus to the driver **3**, the effect of the notification N is sufficiently obtained.

(36) As described above, according to the present embodiment, a notification N of appropriate strength can be given to the driver **3** of the vehicle **1**.

### 3. Examples of Notification Control Process

(37) Examples of the notification control process according to the present embodiment will be described below.

#### 3-1. First Example

(38) FIG. **6** is a flowchart showing a first example of the notification control process.

(39) In step **S100**, the control device **50** determines whether a notification condition is satisfied. In other words, the control device **50** determines whether a notification to the driver **3** is necessary.

(40) For example, the notification condition is that the level of alertness of the driver **3** falls below a threshold. The control device **50** calculates the level of alertness of the driver **3** based on



information detected by the driver monitor **40**. Specifically, the control device **50** detects the face orientation, line of sight, eyes opening degree, mouth opening degree, etc. of the driver **3** by analyzing images acquired by the camera. For example, the level of alertness is calculated so that the smaller the eyes opening degree of the driver **3**, the lower the level of alertness. As another example, the level of alertness is calculated so that the farther away the face orientation of the driver **3** is from the forward direction **DO**, the lower the level of alertness.

(41) As another example, the notification condition is that any sign of drowsiness of the driver **3** is detected. Signs of drowsiness indicate a reduced level of alertness of the driver **3**, and mean a state before and after the driver **3** feels drowsy. Typically, the signs of drowsiness appear as characteristic drowsy behavior before and after the driver **3** feels drowsy. Examples of the drowsy behavior include half-closed eyes, heavy eyelids, frequent blinking, long yawns, and head movements. The control device **50** detects signs of drowsiness of the driver **3** based on images captured by the camera of the driver monitor **40**.

(42) As still another example, the notification condition may be that it becomes necessary to request the driver **3** to monitor the surroundings during autonomous driving of the vehicle **1**. As yet another example, the notification condition may be that it becomes necessary to request the driver **3** to hold the steering wheel during autonomous driving of the vehicle **1**. As a further example, the notification condition may be that it becomes necessary to request the driver **3** to switch to manual driving during autonomous driving of the vehicle **1**. As a still further example, the notification condition may be that it becomes necessary to request the driver **3** to perform a braking operation. As a yet further example, the notification condition may be that the driver assistance control for assisting with driving of the vehicle **1** is activated. These pieces of information are obtained from a vehicle control system that controls the vehicle **1**. The vehicle control system will be described later in Section 5.

(43) When the notification condition is satisfied (step **S100**: YES), a notification **N** to the driver **3** is necessary, and the process proceeds to step **S110**. On the other hand, when the notification condition is not satisfied (step **S100**: NO), no notification **N** to the driver **3** is necessary, and the process in this cycle ends.

(44) In step **S110**, the control device **50** first initializes the target level to the first level **LV1**. The control device **50** then controls the first visual device **20-1** to give the first visual notification **NV1** from the first visual device **20-1**. At this time, the control device **50** does not give any sound notification **NA** from the speaker **30**.

(45) In step **S115**, the control device **50** determines whether the driver **3** has reacted to the first visual notification **NV1** within a certain period of time. For example, the reaction of the driver **3** is that the line of sight of the driver **3** approaches the first direction **D1** to the first visual device **20-1**. The reaction of the driver **3** may be that the line of sight of the driver **3** substantially coincides with the first direction **D1** to the first visual device **20-1**. As a further example, the reaction of the driver **3** may be that the level of alertness of the driver **3** recovers above the threshold. As a still further example, the reaction of the driver **3** may be that the driver **3** performs a requested operation. Examples of the operation that is requested to the driver **3** include holding the steering wheel, switching to manual driving, and operating the brakes. In any case, the control device **50** can detect whether the driver **3** has reacted to the first visual notification **NV1**, based on information obtained by the driver monitor **40**. When the driver **3** has reacted to the first visual notification **NV1** (step **S115**: YES), the process in this cycle ends. On the other hand, when the driver **3** has not reacted to the first visual notification **NV1** (step **S115**: NO), the process proceeds to step **S120**.

(46) In step **S120**, the control device **50** sets (changes) the target level to the second level **LV2**. The control device **50** then controls the second visual device **20-2** to give the second visual notification **NV2** from the second visual device **20-2**. At this time, the control device **50** does not output any sound notification **NA** from the speaker **30**.

(47) In step **S125**, the control device **50** determines whether the driver **3** has reacted to the second

visual notification NV2 within a certain period of time. For example, the reaction of the driver 3 is that the line of sight of the driver 3 approaches the second direction D2 to the second visual device 20-2. The reaction of the driver 3 may be that the line of sight of the driver 3 substantially coincides with the second direction D2 to the second visual device 20-2. As a further example, the reaction of the driver 3 may be that the level of alertness of the driver 3 recovers above the threshold. As a still further example, the reaction of the driver 3 may be that the driver 3 performs a requested operation. The control device 50 can detect whether the driver 3 has reacted to the second visual notification NV2, based on information obtained by the driver monitor 40. When the driver 3 has reacted to the second visual notification NV2 (step S125: YES), the process in this cycle ends. On the other hand, when the driver 3 has not reacted to the second visual notification NV2 (step S125: NO), the process proceeds to step S130.

(48) In step S130, the control device 50 sets (changes) the target level to the third level LV3. The control device 50 then controls the speaker 30 to give a sound notification NA from the speaker 30.

(49) In step S135, the control device 50 determines whether the driver 3 has reacted to the sound notification NA within a certain period of time. For example, the reaction of the driver 3 is that the level of alertness of the driver 3 recovers above the threshold. As another example, the reaction of the driver 3 may be that the driver 3 performs a requested operation. The control device 50 can detect whether the driver 3 has reacted to the sound notification NA, based on information obtained by the driver monitor 40. When the driver 3 has reacted to the sound notification NA (step S135: YES), the process in this cycle ends. On the other hand, when the driver 3 has not reacted to the sound notification NA (step S135: NO), the process proceeds to step S140.

(50) In step S140, the control device 50 performs emergency control. For example, the control device 50 controls the vehicle 1 to stop at a safe place (e.g. on the shoulder).

### 3-2. Second Example

(51) FIG. 7 is a flowchart showing a second example of the notification control process. Step S100 is the same as that in the first example described above. When the notification condition is satisfied (step S100: YES), the process proceeds to step S105.

(52) In step S105, the control device 50 determines (sets) the target level of the strength of a notification N based on the result in step S100. For example, the control device 50 compares the level of alertness of the driver 3 to a first threshold TH1, a second threshold TH2, and a third threshold TH3. The second threshold TH2 is lower than the first threshold TH1, and the third threshold TH3 is lower than the second threshold TH2 ( $TH1 > TH2 > TH3$ ). When the level of alertness is lower than the first threshold TH1 and equal to or higher than the second threshold TH2, the target level is set to the first level LV1. When the level of alertness is lower than the second threshold TH2 and equal to or higher than the third threshold TH3, the target level is set to the second level LV2. When the level of alertness is lower than the third threshold TH3, the target level is set to the third level LV3.

(53) When the target level is the first level LV1, the process proceeds to step S110 shown in FIG. 6. When the target level is the second level LV2, the process proceeds to step S120 shown in FIG. 6. When the target level is the third level LV3, the process proceeds to step S130 shown in FIG. 6.

### 4. Modifications

(54) FIG. 8 is a conceptual diagram showing various patterns of the notification control process.

(55) Pattern (A): When the target level is the first level LV1, the first visual notification NV1 is selected. When the target level is the second level LV2, the second visual notification NV2 is selected. When the target level is the third level LV3, a sound notification NA is selected.

(56) Pattern (B): When the target level is the first level LV1, the first visual notification NV1 is selected. When the target level is the second level LV2 or the third level LV3, the second visual notification NV2 is selected.

(57) Pattern (C): When the target level is the first level LV1 or the second level LV2, the first visual notification NV1 is selected. When the target level is the third level LV3, the second visual

notification NV2 is selected.

(58) Pattern (D): When the target level is the first level LV1, a visual notification NV is selected. The visual notification NV herein may be the first visual notification NV1 or may be the second visual notification NV2. When the target level is the second level LV2 or the third level LV3, a sound notification NA is selected.

(59) Pattern (E): When the target level is the first level LV1 or the second level LV2, a visual notification NV is selected. The visual notification NV herein may be the first visual notification NV1 or may be the second visual notification NV2. When the target level is the third level LV3, a sound notification NA is selected.

(60) The process flow described above in Section 3 is modified as appropriate for each pattern. The technical effects described above can be obtained with any of the patterns.

## 5. Vehicle Control System

### 5-1. Configuration Example

(61) FIG. 9 is a block diagram showing an example of the configuration of a vehicle control system 100 according to the present embodiment. The vehicle control system 100 includes the notification system 10 described above. The vehicle control system 100 includes a sensor group 110, a travel device 120, a Human Machine Interface (HMI) 130, a driver monitor 140, and a control device 150.

(62) The sensor group 110 includes a perception sensor, a vehicle state sensor, a position sensor, etc. The perception sensor perceives (detects) the surroundings of the vehicle 1. Examples of the perception sensor include a camera, a Laser Imaging Detection and Ranging (LIDAR) sensor, and a radar. The vehicle state sensor detects the state of the vehicle 1. The vehicle state sensor includes a speed sensor, an acceleration sensor, a yaw rate sensor, a steering angle sensor, etc. The position sensor detects the position and orientation of the vehicle 1. For example, the position sensor includes a Global Navigation Satellite System (GNSS).

(63) The travel device 120 includes a steering device, a drive device, and a braking device. The steering device steers wheels. For example, the steering device includes an electric power steering (EPS) system. The drive device is a power source that generates a driving force. Examples of the drive device include an engine, an electric motor, and in-wheel motors. The braking device generates a braking force.

(64) The HMI 130 presents various kinds of information to the driver 3 and receives various kinds of information from the driver 3. The HMI 130 includes the visual device 20 and the speaker 30 that are described above. The HMI 130 further includes an input device such as a touch panel.

(65) The driver monitor 140 includes the driver monitor 40 described above. That is, the driver monitor 140 includes a camera for capturing images of the driver 3, a steering touch sensor, etc. The driver monitor 140 may detect driving operations of the driver 3.

(66) The control device 150 is a computer that controls the vehicle 1. The control device 150 includes one or more processors 160 (hereinafter simply referred to as “processor 160”) and one or more storage devices 170 (hereinafter simply referred to as “storage device 170”). The processor 160 performs various processes. For example, the processor 160 includes a central processing unit (CPU). The storage device 170 stores various kinds of information. Examples of the storage device 170 include a volatile memory, a nonvolatile memory, a hard disk drive (HDD), and a solid state drive (SSD). The control device 150 includes the control device 50 of the notification system 10.

(67) A vehicle control program 180 is a computer program that is executed by the processor 160. The functions of the control device 150 are implemented through cooperation between the processor 160 that executes the vehicle control program 180 and the storage device 170. The vehicle control program 180 is stored in the storage device 170. Alternatively, the vehicle control program 180 may be recorded on a computer-readable recording medium.

### 5-2. Driving Environment Information

(68) The control device 150 acquires driving environment information 190 using the sensor group

**110.** The driving environment information **190** indicates the driving environment of the vehicle **1**. The driving environment information **190** is stored in the storage device **170**.

(69) The driving environment information **190** includes surroundings information that indicates the perception results from the perception sensor. For example, the surroundings information includes images captured by a camera. The surroundings information may include object information on objects around the vehicle **1**. Examples of the objects around the vehicle **1** include pedestrians, other vehicles (preceding vehicles, parked vehicles, etc.), white lines, traffic lights, signs, and roadside structures. The object information indicates the relative positions and relative speeds of the objects with respect to the vehicle **1**.

(70) The driving environment information **190** further includes vehicle state information indicating the state of the vehicle **1** detected by the vehicle state sensor.

(71) The driving environment information **190** further includes vehicle position information indicating the position and orientation of the vehicle **1**. The vehicle position information is obtained by the position sensor. Accurate vehicle position information may be acquired by localization using map information and the surroundings information (object information).

(72) The driving environment information **190** further includes information obtained by the driver monitor **140**.

### 5-3. Vehicle Travel Control

(73) The control device **150** performs vehicle travel control for controlling travel of the vehicle **1**. The vehicle travel control includes steering control, drive control, and braking control. The control device **150** performs the vehicle travel control by controlling the travel device **120** (steering device, drive device, and braking device).

(74) The control device **150** may perform autonomous driving control based on the driving environment information **190**. More specifically, the control device **150** generates a travel plan for the vehicle **1** based on the driving environment information **190**. The control device **150** also generates a target trajectory that is necessary for the vehicle **1** to travel according to the travel plan, based on the driving environment information **190**. The target trajectory includes a target position and a target speed. The control device **150** then performs the vehicle travel control so that the vehicle **1** follows the target trajectory.

(75) The control device **150** may perform the driver assistance control for assisting with driving of the vehicle **1**, based on the driving environment information **190**. Examples of the driver assistance control include collision avoidance control and lane departure prevention control. The collision avoidance control controls either or both of steering and deceleration in order to assist with avoidance of a collision between the vehicle **1** and any surrounding object. The lane departure prevention control controls steering in order to reduce the possibility of the vehicle **1** departing from its lane. When a condition to activate the driver assistance control is satisfied, the control device **150** activates the driver assistance control.

(76) The collision avoidance control will be described as an example of the driver assistance control. The control device **150** perceives an object to be avoided in front of the vehicle **1** (e.g., a surrounding vehicle or a pedestrian) based on the surroundings information. The control device **150** also predicts the future positions of both the vehicle **1** and the object to be avoided based on the vehicle state information and the surroundings information, and calculates the possibility of the vehicle **1** colliding with the object to be avoided. A condition to activate the collision avoidance control is that the possibility of the vehicle **1** colliding with the object to be avoided becomes higher than an activation threshold.

(77) The lane departure prevention control will be described as another example of the driver assistance control. For example, when the vehicle **1** drifts toward a lane marking (white line) of its lane, the lane departure prevention control controls steering so that the vehicle **1** returns to the center of its lane. For this purpose, the control device **150** perceives the lane markings of the lane in which the vehicle **1** is traveling based on the surroundings information, and monitors the distances

between the vehicle **1** and the lane markings. A condition to activate the lane departure prevention control is that the distance between the vehicle **1** and one of the lane markings become less than a predetermined distance threshold.

#### 5-4. Notification Control

(78) The control device **150** controls notifications N to the driver **3** by controlling the HMI **130** (visual device **20**, speaker **30**) (see Sections **2** to **4**). More specifically, the control device **150** gives a notification N to the driver **3** when a predetermined notification condition is satisfied.

(79) For example, the notification condition is that the level of alertness of the driver **3** falls below a threshold. The control device **150** can calculate the level of alertness of the driver **3** based on information obtained by the driver monitor **140**.

(80) As another example, the notification condition is that any sign of drowsiness of the driver **3** is detected. The control device **150** can detect signs of drowsiness of the driver **3** based on images captured by the camera of the driver monitor **140**.

(81) As still another example, the notification condition may be that it becomes necessary to request the driver **3** to monitor the surroundings during autonomous driving of the vehicle **1**. As yet another example, the notification condition may be that it becomes necessary to request the driver **3** to hold the steering wheel during autonomous driving of the vehicle **1**. As a further example, the notification condition may be that it becomes necessary to request the driver **3** to switch to manual driving during autonomous driving of the vehicle **1**. As a still further example, the notification condition may be that it becomes necessary to request the driver **3** to perform a braking operation. As a yet further example, the notification condition may be that the driver assistance control is activated.

(82) When the notification condition is satisfied, the control device **150** gives a notification N of appropriate strength corresponding to the target level to the driver **3**.

## Claims

1. A notification method for giving a notification to a driver sitting in a driver's seat of a vehicle, the method comprising: setting a target level of strength of the notification; when the target level is a first level, giving a visual notification from a visual device of the vehicle without giving a sound notification from a speaker of the vehicle; and when the target level is higher than the first level, giving a strong visual notification from the visual device, the strong visual notification being stronger than the visual notification that is given when the target level is the first level, wherein when the target level is the first level: (i) the sound notification is not given from the speaker, and (ii) a first visual notification is given from a first visual device, wherein the visual device comprises the first visual device; when the target level is a second level: (i) the sound notification is not given from the speaker, and (ii) a second visual notification is given from a second visual device, wherein the visual device comprises the second visual device, and the second level being higher than the first level; and a first angle between a direction from the driver's seat toward the first visual device and a forward direction is greater than a second angle between a direction from the driver's seat toward the second visual device and the forward direction.

2. The notification method according to claim 1, wherein: the first angle is greater than a reference angle and smaller than a limit angle, the reference angle being half an angle of a human's average effective visual field, and the limit angle being half an angle of a human's average visual field; and the second angle is smaller than the reference angle.

3. The notification method according to claim 1, wherein when the target level is a third level, the sound notification is given from the speaker, the third level being higher than the second level.

4. The notification method according to claim 3, further comprising: determining whether the notification to the driver is necessary; determining whether the driver reacts to the first visual notification; and determining whether the driver reacts to the second visual notification, wherein

when the notification to the driver is necessary, the target level is initialized to the first level, when the driver does not react to the first visual notification, the target level is set to the second level, and when the driver does not react to the second visual notification, the target level is set to the third level.

5. The notification method according to claim 1, further comprising: determining whether the notification to the driver is necessary; and determining whether the driver reacts to the visual notification when the target level is the first level, wherein when the notification to the driver is necessary, the target level is initialized to the first level, and when the driver does not react to the visual notification, the target level is set to a level higher than the first level.

6. A notification system that gives a notification to a driver sitting in a driver's seat of a vehicle, the notification system comprising one or more processors, wherein: the one or more processors are configured to set a target level of strength of the notification; the one or more processors are configured to, when the target level is a first level, give a visual notification from a visual device of the vehicle without giving a sound notification from a speaker of the vehicle; and the one or more processors are configured to, when the target level is higher than the first level, give a strong visual notification from the visual device, the strong visual notification being stronger than the visual notification that is given when the target level is the first level, wherein when the target level is the first level: (i) the sound notification is not given from the speaker, and (ii) a first visual notification is given from a first visual device, wherein the visual device comprises the first visual device; when the target level is a second level: (i) the sound notification is not given from the speaker, and (ii) a second visual notification is given from a second visual device, wherein the visual device comprises the second visual device, and the second level being higher than the first level; and a first angle between a direction from the driver's seat toward the first visual device and a forward direction is greater than a second angle between a direction from the driver's seat toward the second visual device and the forward direction.

7. A vehicle comprising: a visual device; a speaker; and a control device that is configured to control a notification to a driver sitting in a driver's seat, wherein the control device is configured to set a target level of strength of the notification, the control device is configured to, when the target level is a first level, give a visual notification from the visual device without giving a sound notification from the speaker, and the control device is configured to, when the target level is higher than the first level, give a strong visual notification from the visual device, the strong visual notification being stronger than the visual notification that is given when the target level is the first level, wherein when the target level is the first level: (i) the sound notification is not given from the speaker, and (ii) a first visual notification is given from a first visual device, wherein the visual device comprises the first visual device; when the target level is a second level: (i) the sound notification is not given from the speaker, and (ii) a second visual notification is given from a second visual device, wherein the visual device comprises the second visual device, and the second level being higher than the first level; and a first angle between a direction from the driver's seat toward the first visual device and a forward direction is greater than a second angle between a direction from the driver's seat toward the second visual device and the forward direction.

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